$$= \int_d^R \sqrt{R^2 - x^2}\, dx$$

$a = R$
$x = R\operatorname{sen}\theta$
$dx = R\cos\theta\, d\theta$

$\boxed{\begin{array}{l} \operatorname{sen}^2\theta + \cos^2\theta = 1 \\ \cos^2\theta = 1 - \operatorname{sen}^2\theta \\ \quad a \quad x \end{array}}$

$$= \int \sqrt{R^2 - (R\operatorname{sen}\theta)^2}\, R\cos\theta\, d\theta$$

$$= \int \sqrt{R^2 - R^2\operatorname{sen}^2\theta}\, R\cos\theta\, d\theta$$

$$= \int \sqrt{R^2(1 - \operatorname{sen}^2\theta)}\, R\cos\theta\, d\theta$$

$$= \int \sqrt{R^2(\cos^2\theta)}\, R\cos\theta\, d\theta$$

$$= \int R\cos\theta \cdot R\cos\theta\, d\theta$$

$$= \int R^2\cos^2\theta\, d\theta$$

$$= R^2 \int \cos^2\theta\, d\theta \quad ①$$

$$= R^2 \int \left(\frac{1 + \cos(2\theta)}{2}\right) d\theta \quad ②$$

$$= \frac{R^2}{2}\left[\theta + \frac{\operatorname{sen}(2\theta)}{2}\right] \quad ③$$

$$= \frac{R^2}{2}\left[\theta + \operatorname{sen}\theta\cos\theta\right]$$

① $\boxed{\dfrac{1 + \cos(2\theta)}{2}}$

② $\boxed{\begin{array}{l} = R^2 \displaystyle\int \left(\frac{1 + \cos(2\theta)}{2}\right) d\theta \\ = \dfrac{R^2}{2} \displaystyle\int (1 + \cos 2\theta)\, d\theta \\ = \dfrac{R^2}{2}\left[\theta + \dfrac{\operatorname{sen}2\theta}{2}\right] \end{array}}$

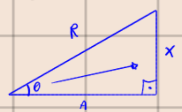
③ $\boxed{\begin{array}{l} \operatorname{sen}(\theta+\theta) = \operatorname{sen}\theta\cos\theta + \operatorname{sen}\theta\cos\theta \\ \operatorname{sen}(2\theta) = 2\operatorname{sen}\theta\cos\theta \\ \dfrac{\operatorname{sen}(2\theta)}{2} = \operatorname{sen}\theta\cos\theta \end{array}}$

VOLTANDO PARA "x", POR MEIO DO TRIÂNGULO RETÂNGULO.

SABEMOS QUE $x = R\operatorname{sen}\theta$

LOGO $\dfrac{x}{R} = \operatorname{sen}\theta = \dfrac{O}{H}$

SEJA "A" O CATETO ADJACENTE A θ.



TEMOS QUE

$$x^2 + A^2 = R^2$$

$$A = \sqrt{R^2 - x^2}$$

TEMOS QUE

$$\cos\theta = \frac{A}{H} = \frac{\sqrt{R^2 - x^2}}{R}$$

$$\theta = \operatorname{arcsen}\left(\frac{x}{R}\right)$$

$$= \frac{R^2}{2}\left[\operatorname{arcsen}\left(\frac{x}{R}\right) + \frac{x}{R}\left(\frac{\sqrt{R^2 - x^2}}{R}\right)\right]$$

$$= \frac{R^2}{2} \cdot \operatorname{arcsen}\left(\frac{x}{R}\right) + \frac{R^2}{2} \cdot \frac{x\sqrt{R^2 - x^2}}{R^2}$$

$$= \frac{R^2}{2} \cdot \operatorname{arcsen}\left(\frac{x}{R}\right) + \frac{x\sqrt{R^2 - x^2}}{2}.$$

AGORA, VOLTAMOS COM O 2 LÁ DO INÍCIO.

$$2\left[\frac{R^2}{2} \cdot \operatorname{arcsen}\left(\frac{x}{R}\right) + \frac{x\sqrt{R^2 - x^2}}{2}\right]_d^R$$

$$2\left\{\left[\frac{R^2}{2} \cdot \operatorname{arcsen}\left(\frac{R}{R}\right) + \frac{R\sqrt{R^2 - R^2}}{2}\right] - \left[\frac{R^2}{2} \cdot \operatorname{arcsen}\left(\frac{d}{R}\right) + \frac{d\sqrt{R^2 - d^2}}{2}\right]\right\}$$

$$2\left[\frac{R^2}{2} \cdot \frac{\pi}{2} - \frac{R^2}{2} \cdot \operatorname{arcsen}\left(\frac{d}{R}\right) - \frac{d\sqrt{R^2 - d^2}}{2}\right]$$

$$2\left[\frac{R^2\pi}{4} - \frac{R^2}{2} \cdot \operatorname{arcsen}\left(\frac{d}{R}\right) - \frac{d\sqrt{R^2 - d^2}}{2}\right]$$

$$\frac{R^2\pi}{2} - R^2\operatorname{arcsen}\left(\frac{d}{R}\right) - d\sqrt{R^2 - d^2} \quad \blacksquare$$